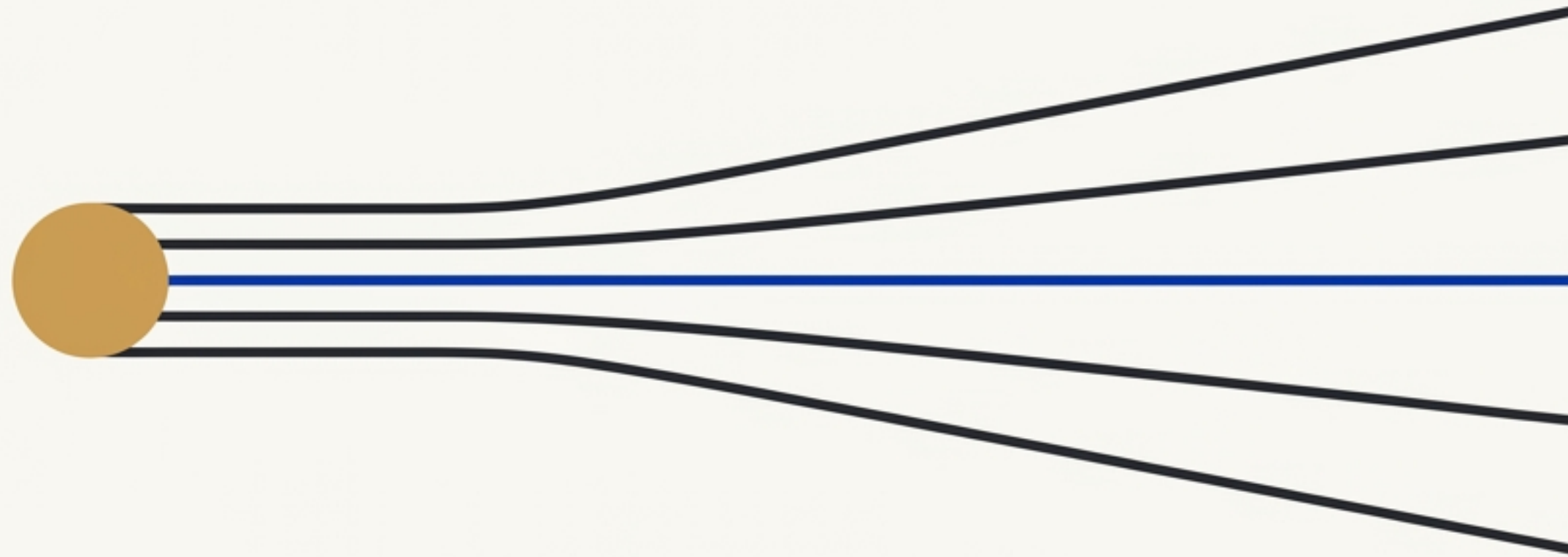


A Voice-Enabled End-to-End Career Copilot

An intelligent agent that scouts jobs, tailors CVs, conducts voice interviews, whispers real-time answers, and offers expert advice.



Kaggle Agents Intensive - Capstone Project
Track: Concierge Agents

The modern job hunt is a fragmented, inefficient, and lonely battle.



Key Insight: Data doesn't flow between stages, creating a disjointed experience where each step starts from scratch.

The process is broken. Job seekers are forced to juggle disparate platforms and manual tasks, leading to lost time, missed opportunities, and a lack of critical feedback.

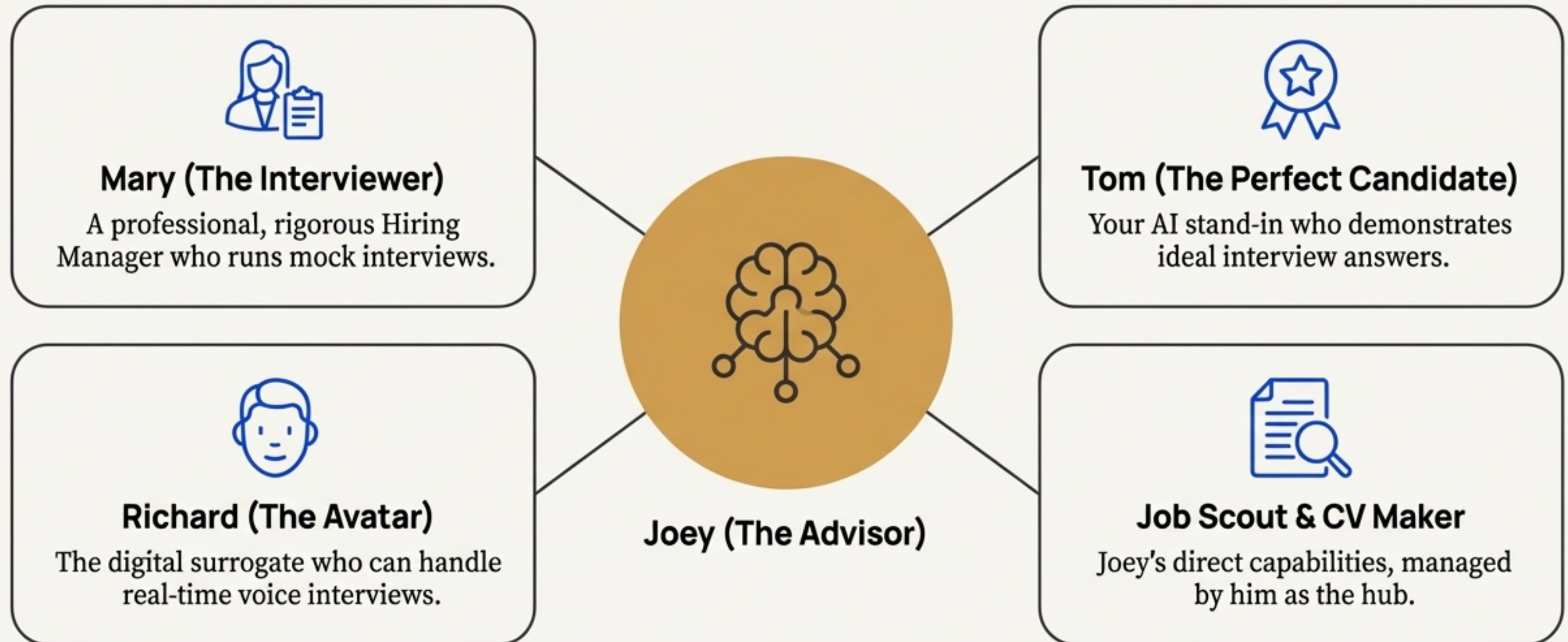
Discovery: Manually sifting through hundreds of irrelevant postings on sites like LinkedIn and Seek.

Preparation: Tediously tailoring resumes and cover letters in Word for each application.

Performance: Practicing interviews in front of a mirror, with no realistic pressure or objective feedback.

Our solution is Joey, an end-to-end Career Copilot powered by a specialist AI team.

Joey is a full-stack career agent built on the Google Agents Development Kit (ADK). He acts as the central hub, managing a team of expert agents who handle every stage of the job search, creating a seamless, data-driven workflow.



Step 1: Joey's Dual-Engine Job Scout automates opportunity discovery.

User Problem

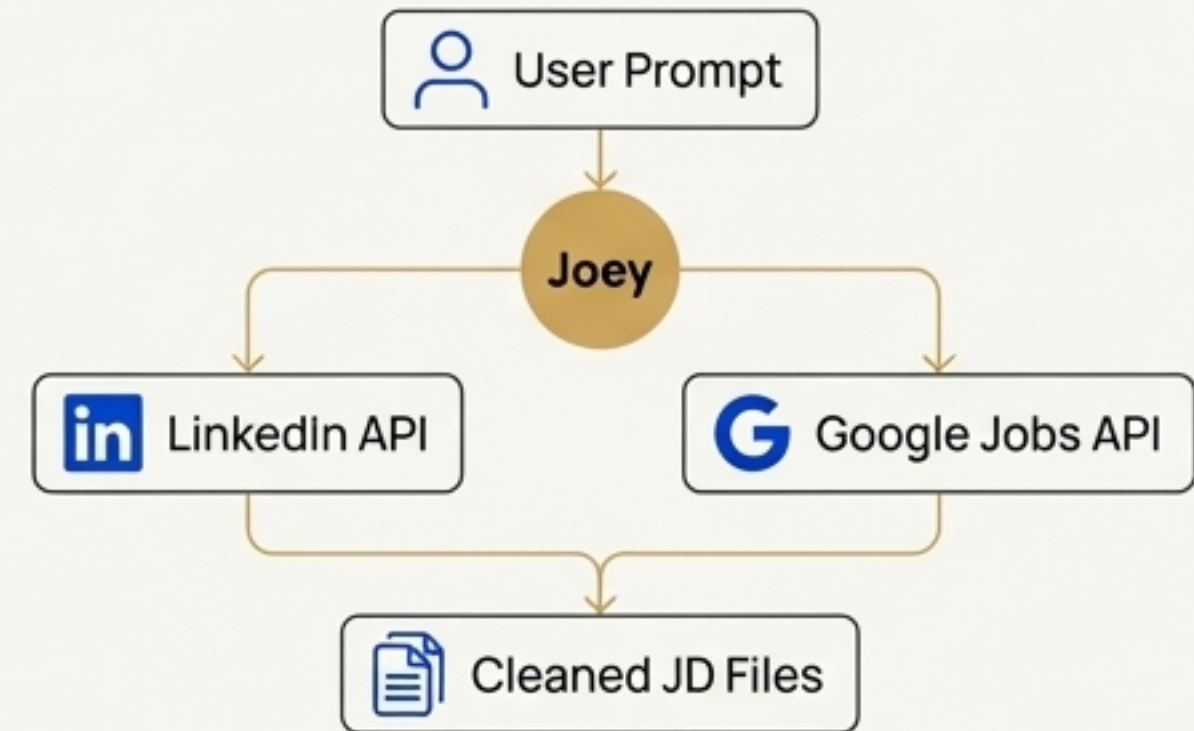
“Manually filtering endless job boards is exhausting and inefficient.”

Joey's Solution

The user issues a simple natural language command. Joey connects to external data sources, to external data sources, fetches relevant positions, cleans the data, and archives it locally.

```
> Joey, find TM1 Developer jobs in Sydney posted this week.
```

Under the Hood



- **Integration:** Connects to LinkedIn & Google Jobs via RapidAPI.
- **Custom Tools:** Utilizes custom function tools within the Google ADK framework.
- **Process:** Fetches raw data, strips HTML and ads, and saves structured Markdown files to a local knowledge base.

Step 2: The CV Maker Service crafts hyper-personalized application materials.

User Problem

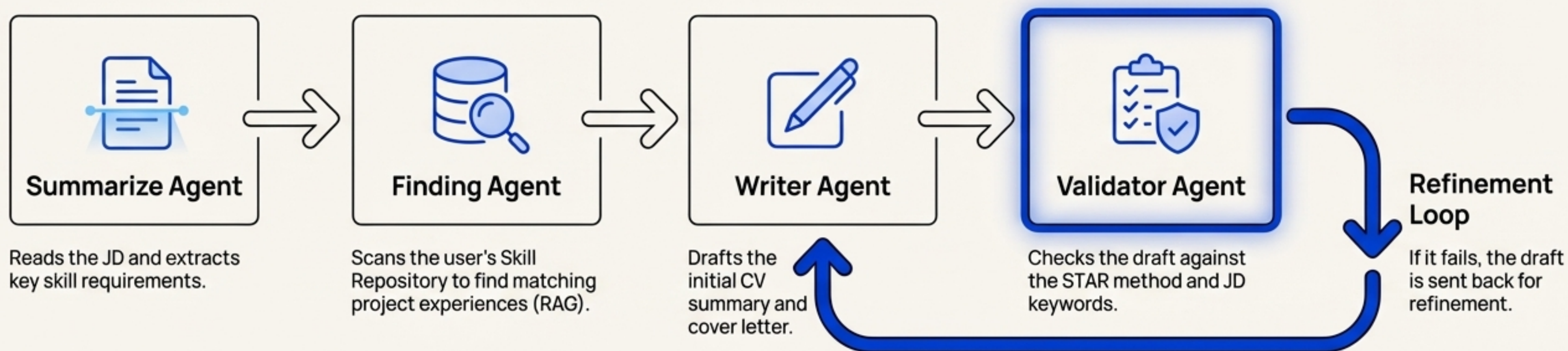
"Creating a unique, tailored CV for every single job is the most time-consuming part of applying."

Joey's Solution

Joey leverages Retrieval-Augmented Generation (RAG) to create bespoke CVs and cover letters. It analyzes the job description and matches its requirements against the user's pre-defined "Skill Repository".

```
> Joey, generate CVs for all downloaded files.
```

The Writer's Room: A Multi-Agent Workflow



Step 3: A suite of simulation tools to master the interview

User Problem: “There’s no way to practice for the pressure and unpredictability of a real interview.”

Joey’s Solution: Joey provides multiple modes of practice, catering to different learning styles.



Prep Mode (The Question Bank)

Don't want to talk? Joey analyzes the JD and generates a comprehensive list of 15-20 likely interview questions for self-study.

> Give me a question bank for the Data Analyst job.



Auto-Simulation (The Observation Deck)

Watch and learn. Joey orchestrates a full interview between Mary (The Interviewer) and Tom (The Perfect Candidate). Tom uses your skills to provide ideal, STAR-based answers.

> Joey, simulate an interview for Reo Group.



Live Mock Interview (The Main Event)

The ultimate test. Engage in a real-time, voice-based interview with Mary, who has only seen your submitted CV and the JD, creating a realistic information gap.

> Interview me for the KPMG position.

Two modes for performance: Your Copilot and Your Avatar.

Interview Copilot (Real-Time Assistance)



Scenario:

You're in an interview (live or mock) and get stuck on a tough question.

Functionality:

You issue a command like "Joey, help me answer this." The system accesses the context of the conversation and your skill repository to generate and whisper a perfectly crafted, conversational answer for you to use.

Value:

Transforms interview anxiety into confidence with a secret expert in your ear.

Avatar Mode (The Autonomous Stand-In)

Scenario:

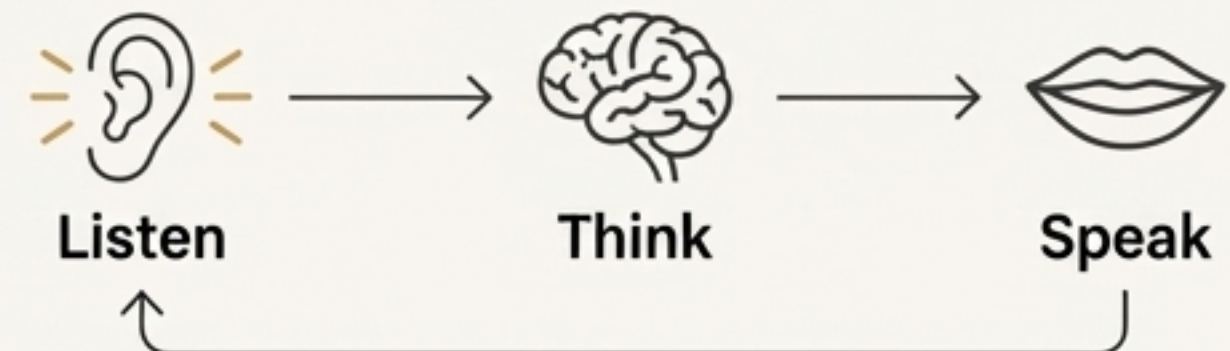
An initial screening call or a high-pressure technical interview.

Functionality:

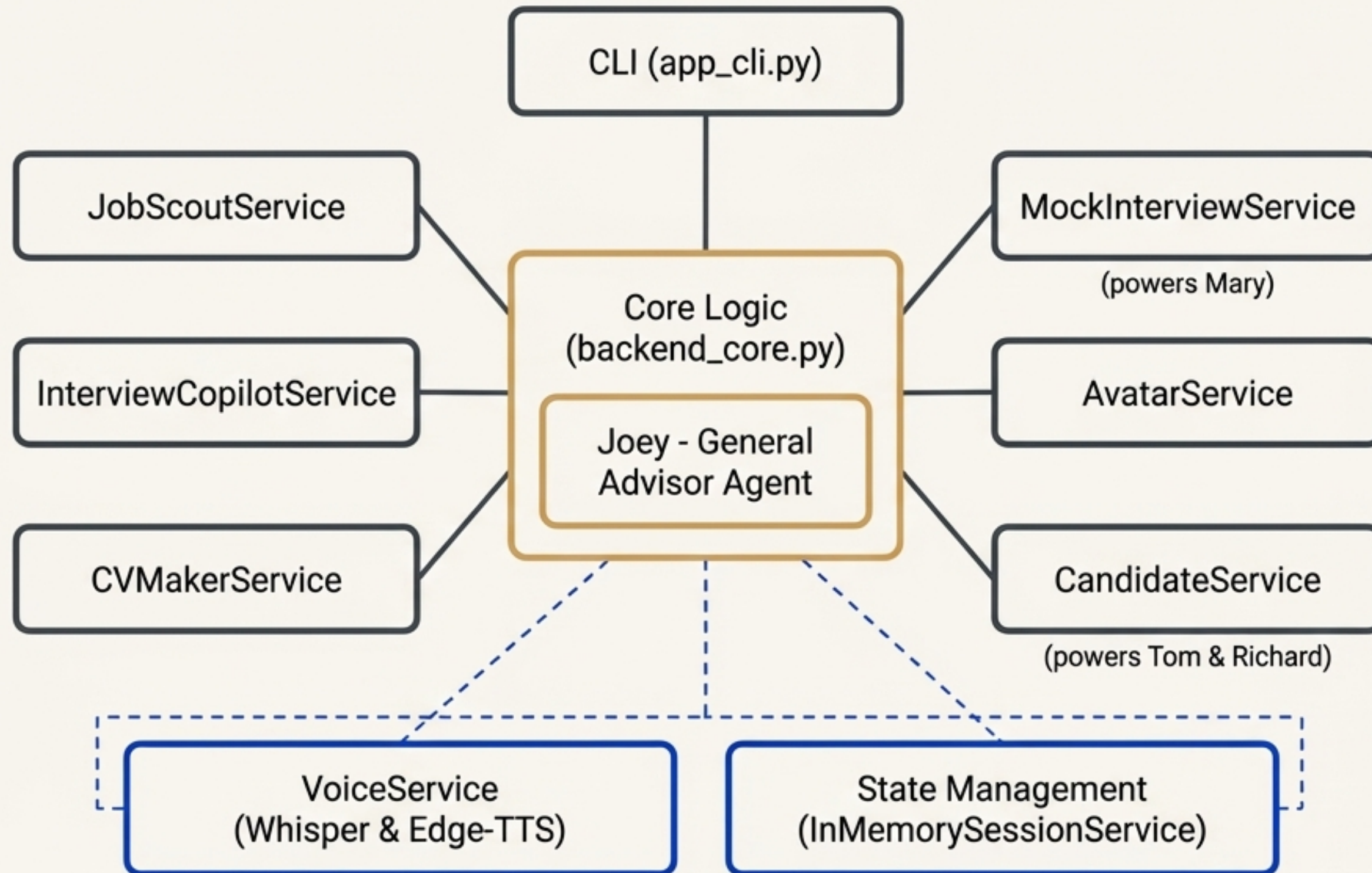
Activate **Richard**, the AI Avatar. He takes over your microphone, listens to the real-world interviewer's questions, and provides autonomous, well-reasoned answers using your voice profile and skill base.



Underlying Logic:



The Hub-and-Spoke Architecture Ties Everything Together.



Key Technologies List

- **Framework:** Google Agents Development Kit (ADK)
- **Reasoning Engine:** Gemini 1.5 Flash
- **Custom Tools:** 8 custom functions (`find_jobs`, `generate_cv`, etc.)
- **State Management:** `InMemorySessionService` + Global App State (`is_interview_active`)
- **Data Privacy:** Fully localized deployment ensures user data (CV, skills, transcripts) remains on their own machine.

Meeting the Capstone Project Requirements

Requirement	Implemented in Joey
Multi-agent System	✓ Yes. Implemented both Sequential agents (in the CV Maker's Writer-Validator workflow) and Parallel agents (Hub dispatching tasks to specialist services).
Custom Tools	✓ Yes. Developed 8 distinct custom function tools including <code>find_jobs</code> , <code>generate_cv</code> , <code>start_interview</code> , and <code>update_profile</code> to give the agent real-world capabilities.
Sessions & Memory	✓ Yes. Utilizes InMemorySessionService to manage session state, crucial for features like locking the system into 'interview mode' and maintaining context.
Long-running Operations	✓ Yes. The mock interview and avatar modes are long-running operations that maintain state, listen for input, and can be paused or resumed by the user.

Securing Bonus Points Through Advanced Implementation.

Effective Use of Gemini

- ✓ **Achieved (5 points):** Gemini 1.5 Flash is not just an add-on; it serves as the core reasoning and generation engine for every agent in generation engine for every agent in the system, from understanding user intent to crafting nuanced interview responses.

Agent Deployment

- ✓ **Achieved (5 points):** The agent has been successfully deployed using Google Cloud Run. Documentation is provided in the repository to reproduce the deployment, demonstrating a production-ready approach.

YouTube Video Submission

- ✓ **Achieved (10 points):** A concise (under 3 min) video has been produced to articulate the problem, demonstrate the solution in action, and explain the architecture.



<https://youtu.be/wMDjGDitthE>

Project Resources & Codebase.

Kaggle Writeup

<https://www.kaggle.com/competitions/agents-intensive-capstone-project/writeups/a-voice-enabled-end-to-end-career-copilot>



GitHub Repository

<https://github.com/smilingsand/Joey-career-copilot>



Video Demonstration

<https://youtu.be/wMDjGDitthE>



The Future of Career Navigation is Collaborative, Not Solitary.

Joey transforms the job search from an isolated struggle into a dynamic partnership with a dedicated team of AI specialists. It's a blueprint for how personalized agents can serve as mentors, researchers, and coaches, empowering individuals to not just find their minis torn to not find their next job, but to strategically manage their entire career.

By automating the toil and simulating the trials, we allow human talent to focus on what matters most: showing up with confidence and telling their own story.